Unit 4 · Session 1 · Tue, Nov 10, 2026

Endocrine Pharmacotherapy — Session 1

3 parts · every fact cited to its slide or textbook page. About 120 minutes of focused study.

Study plan

20 min
Read the guide
One pass, top to bottom. Bold = what the lecture emphasized.

55 min
Diagrams & tables
Redraw each flow from memory; cover a table column and recall it.

10 min
Cram sheet
Hide answers and quiz yourself. Repeat until it's automatic.

30 min
Self-test
Answer before you look. Read every explanation, even when right.

5 min
Objectives check
Tick each objective only if you can teach it out loud.

HOW TO READ THIS GUIDE

- **Bold** = emphasized on the lecture slides
- TB textbook only · SG course study guide only
- S12 = slide 12 · S12 (notes) = speaker notes · SG U1-R3 = study guide item · p. 17 = Katzung page
- Open **Sidekick** (bottom right) to search, listen, drill flashcards, quiz, or chat about this lecture

The whole lecture in 6 sentences

1. One **axis** (hypothalamus → pituitary → gland) with **negative feedback**: every exogenous hormone suppresses the levels above it. S5; SG U4-R1

2. **Acromegaly**: octreotide/lanreotide or pegvisomant (monitor IGF-1); **prolactinoma**: cabergoline; **DI**: desmopressin (hyponatremia); **SIADH**: fluid restriction, then tolvaptan (start in hospital). S6; S8; S10; SG U4-R2; SG U4-R3; SG U4-R4

3. **Levothyroxine**: empty stomach, separate calcium/iron/antacids by 4 h, TSH at 6–8 weeks, start low in older/cardiac patients. S12; S13; SG U4-R5

4. **Methimazole** first line; **PTU** in the first trimester and **thyroid storm** (also blocks T4→T3); fever/sore throat → CBC. S15; SG U4-R6

5. **Thyroid storm**: beta-blocker + PTU first, **iodine after**, steroids too. S16; S18; SG U4-R7

6. **Chronic steroids** suppress CRH/ACTH → **taper**; abrupt stop → adrenal crisis → IV hydrocortisone + fluids. Antagonists: ketoconazole, metyrapone, mifepristone, spironolactone. S22; S23; S27; SG U4-R8; SG U4-R9

Learning objectives

IO = syllabus unit objective · SO = this session's objectives slide.

IO4.1 Identify major pharmacologic classes used in the management of endocrine conditions and describe their mechanisms of action.

Part 1 · Pituitary: GH, Prolactin & ADH
Part 2 · Thyroid & Antithyroid Drugs
Part 3 · Adrenocorticosteroids & Antagonists

IO4.2 Describe the indications, contraindications, adverse effects, and clinically significant interactions associated with medications used to treat endocrine conditions.

Part 1 · Pituitary: GH, Prolactin & ADH
Part 2 · Thyroid & Antithyroid Drugs
Part 3 · Adrenocorticosteroids & Antagonists

IO4.3 Compare pharmacologic agents within and across therapeutic classes to distinguish clinically relevant differences in efficacy, safety, and use.

Part 2 · Thyroid & Antithyroid Drugs
Part 3 · Adrenocorticosteroids & Antagonists

IO4.4 Select appropriate pharmacologic therapy for common endocrine conditions based on clinical presentation and patient-specific factors.

Part 1 · Pituitary: GH, Prolactin & ADH
Part 2 · Thyroid & Antithyroid Drugs
Part 3 · Adrenocorticosteroids & Antagonists

IO4.5 Identify appropriate dosing, administration, and monitoring considerations for medications used in endocrine conditions.

Part 1 · Pituitary: GH, Prolactin & ADH
Part 2 · Thyroid & Antithyroid Drugs
Part 3 · Adrenocorticosteroids & Antagonists

IO4.6 Explain important medication-use and safety information to patients receiving pharmacotherapy for endocrine conditions.

Part 2 · Thyroid & Antithyroid Drugs
Part 3 · Adrenocorticosteroids & Antagonists

SO1 Describe the hypothalamic-pituitary-target gland axis as a shared regulatory template, and identify where drug therapy intervenes at each level.

Part 1 · Pituitary: GH, Prolactin & ADH

SO2 Identify the major agents used for growth hormone disorders, hyperprolactinemia, and disorders of antidiuretic hormone (diabetes insipidus, SIADH).

Part 1 · Pituitary: GH, Prolactin & ADH

SO3 Describe the mechanisms, indications, and monitoring of thyroid hormone replacement and antithyroid drug therapy, including management of thyroid storm.

Part 2 · Thyroid & Antithyroid Drugs

SO4 Compare glucocorticoid and mineralocorticoid pharmacology, and identify agents used to block adrenocortical hormone synthesis or action.

Part 3 · Adrenocorticosteroids & Antagonists

SO5 Identify dosing, administration, and monitoring considerations, and explain key medication-safety counseling points to patients, including HPA-axis suppression and the need to taper chronic glucocorticoids.

Part 2 · Thyroid & Antithyroid Drugs
Part 3 · Adrenocorticosteroids & Antagonists

ABOUT THIS LECTURE

- **PA PHARMACOLOGY COURSE · LECTURE 4 · SESSION 1 OF 2** — Endocrine Pharmacotherapy. **Session 1: Pituitary, Thyroid, and Adrenal Pharmacology | Tuesday, November 10 | 8:30–11:00 AM** S1
- One template runs this whole unit: hypothalamus drives the pituitary, the pituitary drives a target gland, and the gland's hormone feeds back to turn the levels above it down. Every drug today replaces a hormone, blocks one, or interrupts that loop. S1 (notes); S4 (notes)
- Arc: the shared HPA-template concept, then pituitary, thyroid and adrenal in turn. Preview: gonadal hormones, pancreatic/diabetes pharmacology, and parathyroid disorders continue in Session 2 (Thursday, November 12). S2
- Orient to the arc: the shared HPA-template concept, then three organ systems in turn (pituitary, thyroid, adrenal) that all follow the same hypothalamic-pituitary-target gland logic of Figure 1; Session 2 continues with gonadal hormones, diabetes, and parathyroid disorders on Nov 12. S2 (notes)
- Readings: Katzung's Chapters 37–42 are required for Unit 4 (Hypothalamic & Pituitary Hormones; Thyroid & Antithyroid Drugs; Adrenocorticosteroids & Adrenocortical Antagonists; Gonadal Hormones & Inhibitors; Pancreatic Hormones & Glucose-Lowering Drugs; Agents That Affect Bone Mineral Homeostasis). S3; S3 (notes)
- Session 1 covers Chapters 37–39; Session 2 covers Chapters 40–42. Recommended: Pharmacotherapy Principles and Practice (Chisholm-Burns, Schwinghammer, Malone, Kolesar, Lee, 2022, McGraw-Hill; print ISBN 9781260460278, eText ISBN 9781260460285) as supplemental support, not a substitute. S3; S3 (notes)
- Required text: Katzung's Basic and Clinical Pharmacology, 16th Edition (Todd W. Vanderah, 2024/2025, McGraw-Hill Education; ISBN 1260463303 eBook). S3
- **APPENDIX 1** gives **Formulation, dose, indications, adverse effects, contraindications and toxicity** — your study reference: know the dose logic, the one or two serious adverse effects, the contraindication that changes the decision, and the toxicity you would need to recognize. S29; S29 (notes); S30-51
- **Abbreviation key PO = by mouth SC = subcutaneous IM = intramuscular IV = intravenous ER = extended-release ODT = orally disintegrating tablet LAR = long-acting release BID / TID / QID = two / three / four times daily mcg = microgram** S29; S29 (notes)
- **GH = growth hormone IGF-1 = insulin-like growth factor 1 TSH = thyroid-stimulating hormone T4 / T3 = thyroxine / triiodothyronine DI = diabetes insipidus SIADH = syndrome of inappropriate ADH secretion HPA = hypothalamic-pituitary-adrenal** S29; S29 (notes)
- **CBC = complete blood count LFT = liver function tests QT = QT interval on ECG CYP = cytochrome P450 MI = myocardial infarction MRI = magnetic resonance imaging ARB = angiotensin receptor blocker AV = atrioventricular CNS = central nervous system** S29; S29 (notes)

- **Appendix 2: DRUG PROFILE · SUPPORTING** — supporting drugs explain interactions, not doses. Listed for context; not profiled in detail. S52; S52 (notes)
- **Original figures in this deck:** Figures 1, 6 and 9 (The Hypothalamic-Pituitary-Target Gland Axis; Thyroid Hormone Synthesis: Where Antithyroid Drugs Act; Exogenous Corticosteroids and HPA Axis Suppression) were generated by the course director using the AI tool Claude to illustrate standard, non-copyrighted pharmacologic concepts. S53
- They are original diagrams, not reproduced from Katzung's Basic and Clinical Pharmacology or any other copyrighted source. **Mechanism illustrations:** (AI-generated, illustrative, not reproduced from copyrighted sources; schematic and not to scale): somatostatin analogs and pegvisomant (Figure 2, slide 7); dopamine agonists (Figure 3, slide 9); S53; S53 (notes)
- ADH, desmopressin and tolvaptan (Figure 4, slide 10); levothyroxine absorption (Figure 5, slide 15); thyroid storm targets (Figure 7, slide 19); glucocorticoid receptor signaling (Figure 8, slide 25); steroid synthesis and receptor blockade (Figure 10, slide 30). **Reference:** Anthropic. (2026). Claude (Sonnet 5) [Large language model]. https://claude.ai S53

SLIDE DISCREPANCIES

- **Iodine timing in thyroid storm:** slides say at least 1 hour after PTU; the textbook only says not until thioamides have begun. Use the slide timing. S17; p. 732

1 PART 1 · DRUG CLASS

Pituitary: GH, Prolactin & ADH

SO1 SO2 IO4.1 IO4.2 IO4.4 IO4.5

CLASS

- **One regulatory template governs thyroid, adrenal, and gonadal control alike.** S4
- **Figure 1. Original illustrative diagram (not from Katzung): the hypothalamus drives the anterior pituitary, which drives a target endocrine gland, whose peripheral hormone produces end-organ effects and negatively feeds back to suppress the levels above it.** S5
- This is the conceptual spine for the unit — thyroid, adrenal and gonadal pharmacology all map onto this three-level axis; return to it every time a new gland appears. S5 (notes); SG U4-R1
- The peripheral hormone negatively feeds back on the hypothalamus/pituitary above it — which explains why every exogenous hormone therapy in the unit suppresses the levels above it. SG U4-R1

PROTOTYPE + OTHER DRUGS NAMED

- Somatropin (GH deficiency); octreotide, lanreotide, pegvisomant (GH excess); cabergoline, bromocriptine (prolactin); desmopressin (DI); tolvaptan, demeclocycline (SIADH). S6; S8

MECHANISM OF ACTION

- GH axis (Figure 2): pituitary somatotroph releases GH → GH drives liver IGF-1 production → octreotide and lanreotide bind somatostatin receptors and lower GH release. S7; S7 (notes)
- Pegvisomant blocks the GH receptor so IGF-1 falls → acromegaly symptoms improve (monitor IGF-1). GH rises on pegvisomant, so GH is not used to monitor it. S7; S7 (notes)

INDICATIONS

- ● **GH deficiency:** treated with recombinant human growth hormone (somatropin); monitor growth velocity in children and IGF-1 levels in adults. S6; S7; S6 (notes); SG U4-R2
- ● **GH excess (acromegaly):** treated with somatostatin analogs (octreotide, lanreotide — suppress GH release), the GH-receptor antagonist pegvisomant, or dopamine agonists (cabergoline) in select cases. S6; SG U4-R2

- Dopamine normally inhibits prolactin release — dopamine agonists restore that inhibitory tone. Hypothalamic dopamine is the brake on lactotrophs; prolactinoma or drugs (antipsychotics, metoclopramide) lift the brake. S8; S9; S8 (notes); SG U4-R3
- Cabergoline or bromocriptine activate D2 receptors → prolactin release falls and the tumor often shrinks → cycles and fertility return; galactorrhea resolves. S9; S9 (notes)
- ADH (Figure 4): ADH binds the V2 receptor in the collecting duct → aquaporin-2 channels insert → water is reabsorbed and urine concentrates. S10
- Desmopressin boosts this (diabetes insipidus); tolvaptan or demeclocycline blunt it (SIADH) → free water retained in DI therapy, excreted in SIADH therapy. S10; S10 (notes)
- TB Anterior pituitary hormones & their hypothalamic regulators: continuous (not pulsatile) GnRH analogs down-regulate GnRH receptors and suppress LH and FSH. p. 705

- ● **Clinical pearl:** octreotide also has major uses outside the pituitary axis — variceal bleeding and carcinoid syndrome — worth flagging as a cross-system agent. S6; S6 (notes)
- ● **Indication:** octreotide and lanreotide treat acromegaly; octreotide also treats carcinoid syndrome and variceal bleeding. S7
- Hyperprolactinemia: dopamine agonists are first-line — cabergoline, bromocriptine — used for prolactinomas and drug-induced hyperprolactinemia. S8
- Diabetes insipidus (ADH deficiency/resistance): desmopressin (DDAVP) for central DI. SIADH (ADH excess): fluid restriction first-line; vasopressin-receptor antagonists (tolvaptan) or demeclocycline for refractory cases. S8; S10; SG U4-R4
- ● **Somatropin (GH deficiency):** monitor growth velocity or IGF-1. S7

ADVERSE EFFECTS (BOXED WARNINGS FIRST)

- ● **Adverse effects:** GI upset, gallstones and bradycardia with octreotide and lanreotide (also glucose changes). S7; S7 (notes)
- ● **Limitation:** pegvisomant does not shrink the tumor; monitor IGF-1 and liver tests. S7
- ● **Adverse effects:** nausea, orthostatic hypotension and dizziness (dopamine agonists). S9
- Cabergoline is twice weekly and better tolerated; high doses link to valvulopathy. S9 (notes)
- ● **Caution:** high-dose cabergoline in Parkinson patients is linked to cardiac valve fibrosis; use caution with severe psychiatric disease and avoid unmonitored antipsychotic changes. S9; S9 (notes)

CONTRAINDICATIONS

Not covered in materials

- ● **Desmopressin:** risk is hyponatremia; monitor sodium and limit fluids. S10; S10 (notes)
- ● **Tolvaptan:** can overcorrect sodium (osmotic demyelination) and cause liver injury; start in hospital. S10; S10 (notes)
- ● **SIADH:** first-line is fluid restriction; demeclocycline causes photosensitivity and nephrotoxicity. S10
- TB Short stature from GH deficiency, Turner syndrome and other growth disorders is the main somatropin use; long-acting weekly forms (somapacitan) exist. p. 707; p. 721
- TB Growth hormone (somatropin) adverse effects are rare: pseudotumor cerebri, slipped capital femoral epiphysis, scoliosis progression, edema, hyperglycemia. p. 708
- TB Mecasermin (recombinant IGF-I for GH-unresponsive growth failure) causes hypoglycemia — give with a carbohydrate meal within 20 minutes. p. 708
- TB Octreotide: nausea, cramps, steatorrhea, gallstones, and vitamin B12 deficiency with long-term use. p. 709

DRUG INTERACTIONS

- Antipsychotics and metoclopramide raise prolactin by blocking dopamine. S9 (notes); S52

PHARMACOKINETICS

- ● **Dosing:** cabergoline is better tolerated and given twice weekly; bromocriptine is daily. S9; S9 (notes)
- TB Octreotide is 45 times more potent than somatostatin at inhibiting GH; somatostatin analogs come as long-acting monthly depots. Cabergoline's half-life is about 65 hours. p. 709; p. 716

MONITORING

- ● **Monitoring:** pregnancy testing and prolactin monitoring (dopamine agonists). Monitor IGF-1 for acromegaly therapy. S9; S7 (notes)

- **TB** Pegvisomant does not lower GH secretion and may let the adenoma grow — monitor IGF-I and yearly pituitary MRI. p. 709

CLINICAL PEARL / WHY

- DI vs. SIADH is a classic exam pairing — opposite problems; anchor it to "which way does free water move" rather than drug names in isolation. S8; S8 (notes)
- Octreotide reappears in GI/variceal contexts — drug classes are not siloed by organ system. S6 (notes)
- **TB** Hyperprolactinemia causes hypogonadism — infertility, amenorrhea, galactorrhea in women; low libido and ED in men; macroadenomas can compress the optic nerves. p. 715
- **TB** Oxytocin (posterior pituitary) is given IV to start or augment labor; it also triggers milk letdown. Its half-life is about 5 minutes. p. 717

Figure 1 · The hypothalamic-pituitary-target gland axis. Original illustrative diagram (not from Katzung): hypothalamus → anterior pituitary → target gland → peripheral hormone and end-organ effects, with negative feedback to the levels above. Trace the axis and the feedback arrows. S5; S5 (notes)

Figure 2 · Somatostatin analogs and pegvisomant. Schematic (illustrative) — Figure 2. Octreotide and lanreotide lower GH release; pegvisomant blocks the GH receptor so IGF-1 falls; acromegaly improves — monitor IGF-1. S7; S7 (notes)

Figure 3 · Dopamine agonists: restoring the brake on prolactin. Schematic (illustrative) — Figure 3. Dopamine is the brake on prolactin; prolactinomas or dopamine blockers lift it; cabergoline and bromocriptine (D2 agonists) restore it — prolactin falls, cycles and fertility return. S9; S9 (notes)

Figure 4 · ADH, desmopressin and tolvaptan. Schematic (illustrative) — Figure 4. ADH → V2 → aquaporin-2 → water reabsorbed. Desmopressin keeps water in (DI; hyponatremia risk); tolvaptan lets it pass out (SIADH; overcorrection risk). S10; S10 (notes)

Opposite ADH problems (S8; S10)

	Diabetes insipidus	SIADH
Problem	ADH deficiency/resistance — water lost	ADH excess — water retained
Drug	Desmopressin (central DI)	Fluid restriction; tolvaptan or demeclocycline if refractory
Watch for	Hyponatremia — limit fluids	Overcorrection (osmotic demyelination); liver injury; photosensitivity

How to tell them apart. Sources: S8; S10; SG U4-R4

GH-axis profiles: formulation, dose, indications

Drug	Class	Formulation	Adult dose	Indications
Somatropin (Genotropin, Norditropin)	**Recombinant human growth hormone**	SC injection pens and cartridges (several mg strengths)	Adults: start about 0.2 mg SC daily and titrate by IGF-1; children: weight-based daily SC dosing	GH deficiency (children and adults), Turner syndrome, other growth disorders

Drug	Class	Formulation	Adult dose	Indications
Octreotide (Sandostatin)	**Somatostatin analog**	SC injection 50/100/500 mcg/mL; LAR depot IM 10/20/30 mg; oral capsule	Acromegaly: 50 mcg SC TID, titrate to 100-200 mcg TID; LAR 20 mg IM every 4 weeks	Acromegaly, carcinoid syndrome, VIPoma, esophageal variceal bleeding
Lanreotide (Somatuline Depot)	**Somatostatin analog**	Deep SC injection 60/90/120 mg prefilled syringe	90 mg deep SC every 4 weeks for 3 months, then adjust between 60 and 120 mg	Acromegaly, neuroendocrine tumors, carcinoid syndrome
Pegvisomant (Somavert)	**GH receptor antagonist**	SC injection vials 10/15/20/25/30 mg	Load 40 mg SC, then 10 mg SC daily, titrate by IGF-1 (max 30 mg/day)	Acromegaly not controlled by surgery or somatostatin analogs

How to tell them apart. Sources: S30; S31; S32; S33

GH-axis profiles: adverse effects, contraindications, toxicity

Drug	Adverse effects	Contraindications	Toxicity / overdose
Somatropin (Genotropin, Norditropin)	Injection-site reaction, edema, joint pain, headache, hyperglycemia	Active malignancy, closed epiphyses (for growth promotion), acute critical illness	Fluid retention, glucose intolerance, intracranial hypertension, slipped capital femoral epiphysis; titrate by IGF-1
Octreotide (Sandostatin)	GI upset, diarrhea, gallstones, bradycardia, glucose changes	Hypersensitivity	Gallstones, hypo- or hyperglycemia, bradycardia and conduction block; monitor IGF-1, glucose, gallbladder
Lanreotide (Somatuline Depot)	Diarrhea, abdominal pain, gallstones, injection-site reaction, bradycardia	Hypersensitivity	Gallstones, glucose dysregulation, bradycardia; monitor GH and IGF-1
Pegvisomant (Somavert)	Injection-site reaction, headache, raised liver enzymes, flu-like symptoms	Hypersensitivity	Liver injury; monitor tumor size by MRI; GH levels rise and are not used for monitoring

How to tell them apart. Sources: S30; S31; S32; S33

Prolactin and ADH profiles: formulation, dose, indications

Drug	Class	Formulation	Adult dose	Indications
Cabergoline (Dostinex)	**Dopamine D2 agonist (ergot-derived)**	Tablet 0.5 mg	0.25 mg PO twice weekly, increase by 0.25 mg every 4 weeks to usual max 1 mg twice weekly	Hyperprolactinemia, prolactinoma
Bromocriptine (Parlodel, Cycloset)	**Dopamine D2 agonist (ergot-derived)**	Tablet 2.5 mg; capsule 5 mg; Cycloset 0.8 mg tablet	1.25-2.5 mg PO at bedtime, titrate to 2.5 mg BID-TID for prolactinoma (max 15 mg/day)	Hyperprolactinemia, acromegaly (adjunct), Parkinson disease, type 2 diabetes (Cycloset)
Desmopressin (DDAVP)	**Synthetic vasopressin V2 agonist**	Tablet 0.1/0.2 mg; nasal spray; SC/IV 4 mcg/mL; ODT 60 mcg	Central DI: 0.05-0.2 mg PO BID, or 10-40 mcg intranasal daily in divided doses; titrate to urine output	Central diabetes insipidus, nocturnal enuresis, hemophilia A and von Willebrand disease (IV)
Tolvaptan (Samsca)	**Vasopressin V2 receptor antagonist**	Tablet 15/30 mg	15 mg PO once daily, titrate to 30-60 mg; start in hospital	Hyponatremia from SIADH or heart failure
Demeclocycline	**Tetracycline with ADH-antagonist effect**	Tablet 150/300 mg	150-300 mg PO 3-4 times daily for SIADH (off-label)	Refractory SIADH (off-label), bacterial infections

How to tell them apart. Sources: S34; S35; S36; S37; S38

Prolactin and ADH profiles: adverse effects, contraindications, toxicity

Drug	Adverse effects	Contraindications	Toxicity / overdose
Cabergoline (Dostinex)	Nausea, dizziness, headache, orthostatic hypotension	Uncontrolled hypertension, cardiac valvulopathy, ergot hypersensitivity	Cardiac valve fibrosis at high doses; impulse-control and psychiatric effects; overdose causes hypotension and nausea
Bromocriptine (Parlodel, Cycloset)	Nausea, orthostatic hypotension, dizziness, headache, nasal congestion	Uncontrolled hypertension, pregnancy-related hypertension, ergot hypersensitivity	Hypotension and syncope, psychosis, fibrosis with long-term high doses; take with food
Desmopressin (DDAVP)	Headache, nausea, flushing, hyponatremia	Hyponatremia history, polydipsia, eGFR below 50, heart failure	Water intoxication with severe hyponatremia and seizures; limit fluids and monitor sodium
Tolvaptan (Samsca)	Thirst, dry mouth, polyuria, hypernatremia, liver injury	Urgent need to raise sodium, hypovolemia, anuria, strong CYP3A4 inhibitors	Overly rapid sodium correction (osmotic demyelination); liver injury; limit therapy to 30 days

Drug	Adverse effects	Contraindications	Toxicity / overdose
Demeclocycline	Photosensitivity, nausea, nephrogenic diabetes insipidus	Pregnancy, children under 8 years, hypersensitivity, severe renal impairment	Nephrotoxicity, photosensitivity, tooth discoloration; separate from calcium and antacids

How to tell them apart. Sources: S34; S35; S36; S37; S38

WHY IT HAPPENS

- Why is GH useless for monitoring pegvisomant? The drug blocks the receptor, not the release — GH actually rises; IGF-1 is what falls. S7 (notes)

HIGH-YIELD

- Acromegaly: octreotide/lanreotide or pegvisomant (monitor IGF-1); prolactinoma: cabergoline; DI: desmopressin (hyponatremia); SIADH: fluid restriction, then tolvaptan. S6; S8; S10

2 PART 2 · DRUG CLASS

Thyroid & Antithyroid Drugs

SO3 SO5 IO4.1 IO4.2 IO4.3 IO4.4 IO4.5 IO4.6

CLASS

- **Replacing a deficient gland, or slowing down an overactive one.** S11
- **Figure 6. Original illustrative diagram (not from Katzung): iodide trapping, organification, and coupling produce T3/T4, which undergo peripheral T4-to-T3 conversion; methimazole and PTU block organification and coupling, while PTU additionally blocks peripheral conversion.** S14
- Figure 6 is the key to the next slides: PTU, not methimazole, is preferred in thyroid storm because it blocks an extra step. S14 (notes)

PROTOTYPE + OTHER DRUGS NAMED

- Levothyroxine (replacement); methimazole, propylthiouracil (thionamides); propranolol; potassium iodide. S12; S15; S16

MECHANISM OF ACTION

- **Mechanism: synthetic T4, converted peripherally to active T3.** S12
- Levothyroxine path (Figure 5): tablet taken on an empty stomach → absorbed in the small intestine; calcium, iron and antacid binders block it. S13
- T4 converted to active T3 in tissues → T3 binds nuclear receptors and restores metabolism → TSH falls to the target range (recheck at 6–8 weeks). S13
- Methimazole and PTU block organification and coupling; PTU also blocks peripheral T4→T3 conversion — a faster symptomatic effect in crisis. S14; S15
- Storm targets (Figure 7): Graves thyroid makes excess hormone — PTU blocks synthesis; hormone is released — iodine (after PTU) blocks release. S17

INDICATIONS

- Levothyroxine: first-line for hypothyroidism; narrow therapeutic index — dose titrated using TSH (primary hypothyroidism) with repeat labs every 6–8 weeks after a dose change. S12; S12 (notes)
- About 1.6 mcg/kg/day is full replacement; estrogen raises needs. ● **Pregnancy:** often raises dose needs; monitor TSH closely. S12 (notes); S13
- Methimazole: first-line for most non-pregnant patients — once-daily dosing, more convenient; avoid in the first trimester of pregnancy (teratogenic risk). S15; S15 (notes)

- T4 converts to T3 in tissues — PTU, propranolol and steroids block conversion; T3 stimulates beta receptors — propranolol blunts tachycardia and tremor; crisis settles with fluids, cooling and support. S17; S17 (notes)
- TB Thyroid physiology: large doses of iodine inhibit organification (Wolff-Chaikoff block); amiodarone, contrast, β blockers and corticosteroids inhibit T4-to-T3 conversion. p. 724; p. 725

- PTU: preferred in the first trimester of pregnancy and in thyroid storm. PTU first trimester (lower teratogenic risk) and thyroid storm (also blocks peripheral T4→T3 conversion). S15; SG U4-R6
- **A life-threatening hyperthyroid crisis requiring rapid, multi-pronged therapy.** Thyroid storm is treated with four drugs that hit four targets. S16; S16 (notes)
- Beta-blockers (propranolol) — control adrenergic symptoms (tachycardia, tremor) and blunt peripheral T4-to-T3 conversion. PTU (preferred over methimazole) — blocks new hormone synthesis and peripheral conversion. S16
- Iodine solution (given after PTU/methimazole) — acutely blocks hormone release; must follow the antithyroid drug to avoid providing substrate for new synthesis. Corticosteroids — blunt peripheral conversion and treat relative adrenal insufficiency. S16
- TB Laboratory diagnosis of Graves disease: suppressed TSH with high free thyroid hormone. Management of Graves disease: antithyroid drugs, radioactive iodine, or thyroidectomy; methimazole preferred except in pregnancy and storm. p. 734
- TB Pregnant hypothyroid patients often need about 25–30% more levothyroxine; add two extra doses a week as soon as pregnancy is confirmed. p. 734

ADVERSE EFFECTS (BOXED WARNINGS FIRST)

- Methimazole boxed warning: agranulocytosis, hepatotoxicity. PTU boxed warning: severe hepatotoxicity (more than methimazole) — reserve for specific indications. S15; S17

CONTRAINDICATIONS

- **Older adults and patients with cardiac disease: start at lower doses and titrate slowly to avoid precipitating angina/arrhythmia.** S12; S13; S12 (notes)
- ● **Propranolol:** avoided in decompensated heart failure and asthma. S17
- TB Radioactive iodine (131I) is avoided in pregnancy and breastfeeding — it destroys the fetal thyroid. p. 732

- TB Thioamide agranulocytosis occurs in 0.1–0.5%, usually within the first 90 days; PTU's severe hepatitis (boxed warning) restricts it to the first trimester, thyroid storm, or methimazole reactions. p. 730; p. 731
- Both thionamides carry agranulocytosis and liver warnings; PTU has the greater liver risk. Fever or sore throat on a thionamide means stop and check a CBC. S15 (notes)
- ● **Over-replacement:** tachycardia, atrial fibrillation and bone loss. S13; S12 (notes)
- TB Iodides given alone can escape in 2–8 weeks and worsen thyrotoxicosis; avoid chronic iodide in pregnancy (fetal goiter). p. 732

DRUG INTERACTIONS

- **Significant absorption interactions: calcium, iron, and antacids can all reduce absorption — separate dosing by at least 4 hours.** S12; SG U4-R5
- ● **Timing:** separate from calcium, iron, antacids, PPIs and bile acid binders by 4 hours. S13
- Amiodarone and lithium can cause hypo- or hyperthyroidism. Estrogens raise thyroid-binding globulin, so levothyroxine needs rise. Rifampin, phenytoin, carbamazepine speed clearance of levothyroxine and steroids. S52; S52 (notes)
- ● **Steroids:** also treat relative adrenal insufficiency; avoid aspirin in thyroid storm (it frees bound hormone). S17

PHARMACOKINETICS

- Take on an empty stomach, 30–60 minutes before breakfast, consistently relative to other medications and food. S12; S13 (notes)
- TB T4 has a 7-day half-life (allows once-daily dosing); T3 about 1 day. Enzyme inducers (rifampin, phenytoin, carbamazepine) speed thyroid hormone metabolism. p. 725; p. 726; p. 733

MONITORING

- ● **Start low:** in older adults and in cardiac disease. Check TSH at 6–8 weeks. S13; S12 (notes)

CLINICAL PEARL / WHY

- The absorption-interaction list is extremely high-yield — many patients also take calcium or iron, so timing counseling matters as much as the drug itself. S12 (notes)

- The pregnancy-trimester switch (PTU in first trimester, methimazole after) is a frequently tested nuance — worth stating explicitly rather than assuming one drug is simply "better." S15 (notes)
- ● **Sequence:** start PTU and beta-blocker first; give iodine at least an hour later. ● **PTU:** boxed warning for hepatotoxicity. S17
- Case: fever, tachycardia, and altered mental status in untreated Graves disease. **What is the correct sequence of pharmacologic therapy, and why does the order matter?** S18
- **Beta-blockade and PTU are started first; iodine solution is given only after PTU, since giving iodine first would supply additional substrate for thyroid hormone synthesis before the gland's synthetic machinery is blocked — potentially worsening the crisis.** S18; S16 (notes); SG U4-R7
- Iodine-first sequencing is the classic high-yield detail: giving iodine first can worsen the crisis by supplying substrate before synthesis is blocked. S16 (notes); S18 (notes)
- TB Subclinical hypothyroidism (high TSH, normal hormones) affects 4–10%; amiodarone causes hypo- or hyperthyroidism (about 3% thyrotoxicosis) that persists after stopping. p. 734; p. 736
- TB Subacute thyroiditis releases stored hormone transiently; nontoxic goiter in the US is most often Hashimoto thyroiditis; neonatal Graves comes from maternal antibodies. p. 735; p. 736

Step by step

Thyroid storm sequence S16; S17; S18

Recognize: fever, tachycardia, altered mental status in Graves →

Propranolol (avoid in asthma/decompensated HF) →

PTU: blocks synthesis and T4→T3 → Iodine at least 1 hour after PTU →

Steroids; fluids, cooling; no aspirin

Figure 5 · Levothyroxine: absorption, conversion and timing. Schematic (illustrative) — Figure 5. Red X marks absorption blocked by calcium, iron or antacids. Blue steps show T4 to T3 conversion and receptor action. Red outcome: TSH falls to target. S13; S13 (notes)

Figure 6 · Thyroid hormone synthesis: where antithyroid drugs act. Original illustrative diagram (not from Katzung): iodide trapping → organification → coupling → T3/T4 release → peripheral T4-to-T3 conversion. Methimazole and PTU block organification and coupling; PTU also blocks conversion. S14; S14 (notes)

Figure 7 · Thyroid storm: four drugs, four targets. Schematic (illustrative) — Figure 7. Red bars mark where PTU (synthesis), iodine (release, after PTU), propranolol and steroids (conversion, symptoms) act. S17; S17 (notes)

Methimazole vs. propylthiouracil (S15)

	Methimazole	Propylthiouracil (PTU)
Use	First-line for most non-pregnant patients; once daily	First trimester of pregnancy; thyroid storm
Extra action	—	Also blocks peripheral T4→T3 conversion
Pregnancy	Avoid in first trimester (teratogenic)	Preferred in first trimester
Boxed warning	Agranulocytosis, hepatotoxicity	Severe hepatotoxicity (more than methimazole)

How to tell them apart. Sources: S15; SG U4-R6

Thyroid drug profiles: formulation, dose, indications

Drug	Class	Formulation	Adult dose	Indications
Levothyroxine (Synthroid, Levoxyl)	**Synthetic thyroid hormone (T4)**	Tablet 25-300 mcg (12 strengths); IV 100-500 mcg vial; liquid and capsule	Full replacement about 1.6 mcg/kg/day PO; start 12.5-50 mcg in older adults or heart disease; recheck TSH in 6-8 weeks	Hypothyroidism, myxedema coma (IV), TSH suppression in thyroid cancer

Drug	Class	Formulation	Adult dose	Indications
Methimazole (Tapazole)	**Thionamide antithyroid drug**	Tablet 5/10 mg	Graves disease: start 10-30 mg PO once daily; adjust to free T4 every 4-6 weeks	Graves disease, toxic nodular goiter, pre-operative control
Propylthiouracil (PTU)	**Thionamide antithyroid drug (also blocks T4 to T3)**	Tablet 50 mg	Start 100 mg PO TID (up to 300-400 mg/day); thyroid storm: 500-1,000 mg load, then 250 mg every 4 hours	First-trimester Graves disease, thyroid storm, methimazole intolerance
Propranolol (Inderal)	**Nonselective beta-blocker**	Tablet 10/20/40/60/80 mg; ER capsule; IV 1 mg/mL	Thyrotoxic symptoms: 20-40 mg PO every 6-8 hours; thyroid storm: 60-80 mg PO every 4 hours or 0.5-1 mg IV	Thyrotoxic symptoms, thyroid storm, hypertension, tremor
Potassium iodide (SSKI, Lugol solution)	**Iodine (blocks hormone release)**	SSKI 1 g/mL solution; Lugol 5% solution; tablets 65/130 mg	Thyroid storm: 5 drops (about 250 mg) SSKI PO every 6 hours, at least 1 hour after a thionamide	Thyroid storm, pre-thyroidectomy preparation, radiation protection

How to tell them apart. Sources: S39; S40; S41; S42; S43

Thyroid drug profiles: adverse effects, contraindications, toxicity

Drug	Adverse effects	Contraindications	Toxicity / overdose
Levothyroxine (Synthroid, Levoxyl)	Usually none at the right dose; over-replacement causes tachycardia, tremor, insomnia, bone loss	Uncorrected adrenal insufficiency, acute MI, hypersensitivity	Over-replacement: atrial fibrillation, angina, osteoporosis; separate from calcium, iron and antacids by 4 hours
Methimazole (Tapazole)	Rash, itching, joint pain, taste change; agranulocytosis (rare)	First trimester of pregnancy; prior methimazole agranulocytosis	Agranulocytosis (fever or sore throat: stop and check CBC), hepatotoxicity, teratogenicity
Propylthiouracil (PTU)	Rash, nausea, taste change, joint pain	Prior PTU liver injury; avoid in children	Boxed warning: severe liver failure; agranulocytosis; ANCA-positive vasculitis
Propranolol (Inderal)	Bradycardia, fatigue, bronchospasm, masks hypoglycemia	Asthma, decompensated heart failure, high-grade AV block, severe bradycardia	Bradycardia, hypotension, bronchospasm; avoid abrupt stopping; may mask hypoglycemia

Drug	Adverse effects	Contraindications	Toxicity / overdose
Potassium iodide (SSKI, Lugol solution)	Metallic taste, GI upset, salivary gland swelling, rash	Giving before a thionamide; iodine allergy	Iodism (rash, swelling, fever); thyroid escape after about 10 days; hypo- or hyperthyroidism

How to tell them apart. Sources: S39; S40; S41; S42; S43

MNEMONIC (MEMORY AID)

- Storm order: Block (beta-blocker), then Block synthesis (PTU), then Block release (iodine) — iodine never first. S16; S18

HIGH-YIELD

- Levothyroxine: empty stomach, separate calcium/iron/antacids by 4 h, TSH at 6–8 weeks, start low in older/cardiac patients. S12; S13; SG U4-R5
- PTU for storm and first trimester; methimazole otherwise; fever/sore throat → stop, CBC. S15; SG U4-R6

3 PART 3 · DRUG CLASS

Adrenocorticosteroids & Antagonists

SO4 SO5 IO4.1 IO4.2 IO4.3 IO4.4 IO4.5 IO4.6

CLASS

- **Glucocorticoid power and mineralocorticoid balance — and what happens when therapy stops.** S19
- **Figure 9. Original illustrative diagram (not from Katzung): chronic exogenous glucocorticoid use suppresses CRH and ACTH via negative feedback, reducing endogenous cortisol production — abrupt withdrawal can precipitate adrenal insufficiency.** S22
- This is the single most clinically important concept in the glucocorticoid section — it explains why tapering (not stopping) is the standard of care. S22 (notes)

PROTOTYPE + OTHER DRUGS NAMED

- Prednisone, dexamethasone, hydrocortisone (glucocorticoids); fludrocortisone (mineralocorticoid); ketoconazole, metyrapone, mifepristone, spironolactone (antagonists). S20; S23

MECHANISM OF ACTION

- Glucocorticoids (e.g., prednisone, dexamethasone): anti-inflammatory and immunosuppressive effects, with a spectrum of relative potency and mineralocorticoid activity across agents. S20
- Signaling (Figure 8): prednisone or cortisol crosses the cell membrane → binds the cytoplasmic glucocorticoid receptor → the complex moves into the nucleus → switches on anti-inflammatory genes and represses NF-kB cytokines → fewer inflammatory cells and mediators. S21; S21 (notes)
- Mineralocorticoids (fludrocortisone): promote renal sodium retention and potassium excretion. S20

INDICATIONS

- Glucocorticoids are used across nearly every organ system for inflammatory/autoimmune disease. S20
- Fludrocortisone: replacement therapy in primary adrenal insufficiency (Addison disease), alongside a glucocorticoid — primary adrenal insufficiency needs BOTH hormone classes replaced, not just cortisol. S20; S20 (notes)
- ● **Ketoconazole:** inhibits multiple steps in steroidogenesis (CYP enzymes); used for Cushing syndrome; significant drug-interaction and hepatotoxicity concerns. S23; SG U4-R9
- ● **Metyrapone:** blocks the final step of cortisol synthesis (11-β-hydroxylase); used diagnostically and for Cushing syndrome. S23; SG U4-R9

- Synthesis chain (Figure 10): cholesterol enters the adrenal cortex → CYP enzymes convert it to cortisol and aldosterone → ketoconazole blocks several enzymes; metyrapone blocks the last cortisol step (11-beta-hydroxylase). S24
- Then cortisol or aldosterone binds its receptor → mifepristone blocks the glucocorticoid receptor; spironolactone blocks the mineralocorticoid receptor. Receptor blockers leave hormone levels high. S24; S24 (notes)

- ● **Mifepristone:** a glucocorticoid-receptor antagonist; used for hyperglycemia associated with Cushing syndrome when surgery is not an option (distinct from its progesterone-receptor antagonist use in pregnancy termination). S23; S23 (notes); SG U4-R9
- ● **Spironolactone:** a mineralocorticoid-receptor antagonist; used for hyperaldosteronism and as a potassium-sparing diuretic — ties back to the diuretic pharmacology concepts from earlier coursework. S23; SG U4-R9
- **TB** Addison disease: hydrocortisone 20–30 mg daily (more during stress) plus fludrocortisone 0.05–0.2 mg/d. p. 747
- **TB** Dexamethasone suppression test diagnoses Cushing syndrome; antenatal glucocorticoids reduce neonatal respiratory distress. p. 748

ADVERSE EFFECTS (BOXED WARNINGS FIRST)

- ● **Adverse effects:** hyperglycemia, osteoporosis, infection, mood change, cataracts, adrenal suppression. S21; S20 (notes)
- ● **Ketoconazole:** hepatotoxicity, QT prolongation, strong CYP3A4 inhibition. S24; S23 (notes)
- ● **Metyrapone:** raises androgens and deoxycorticosterone (hirsutism, hypertension). S24; S23 (notes)
- ● **Mifepristone:** contraindicated in pregnancy; causes hypokalemia; cortisol levels rise, so monitor clinically. ● **Spironolactone:** hyperkalemia and gynecomastia. S24; S24 (notes)
- **TB** Adrenal suppression can occur after more than 2 weeks of corticosteroids; ACTH does not speed recovery. p. 750

CONTRAINDICATIONS

Not covered in materials

- TB Long-term steroids: Cushingoid changes, myopathy, osteoporosis, peptic ulcer, masked infection, hypomania or psychosis, posterior subcapsular cataracts. p. 745; p. 750

DRUG INTERACTIONS

- Azole antifungals raise corticosteroid exposure through CYP3A4. ACE inhibitors, ARBs and potassium add to spironolactone hyperkalemia. Insulin and oral agents need adjusting during steroid therapy. S52; S52 (notes)

PHARMACOKINETICS

- ● **Potency vs. hydrocortisone:** prednisone about 4x, dexamethasone about 25-30x. ● **Mineralocorticoid action:** fludrocortisone has it; hydrocortisone has both. S21; S20 (notes)
- ● **Dosing:** take in the morning with food; consider bone and GI protection. S21
- TB Synthetic corticosteroids — pharmacokinetics & pharmacodynamics: potency and salt-retaining activity differ by agent (equivalent-dose table). p. 746; p. 747
- TB Selection of drug & dosage schedule: give steroids as a single morning dose; alternate-day dosing lessens suppression when disease control allows. p. 750; p. 751

MONITORING

- Fludrocortisone: monitor blood pressure and potassium. S20
- Chronic steroids suppress CRH and ACTH — abrupt withdrawal risks adrenal crisis; taper. Chronic suppression leaves the axis unable to resume adequate cortisol quickly — taper, don't stop abruptly. S22 (notes); SG U4-R8

CLINICAL PEARL / WHY

- Case: chronic high-dose prednisone for a rheumatologic condition, stopped abruptly after running out of refills; now hypotension, weakness, nausea, confusion. **What is the most likely cause of this presentation, and how should it be managed?** S27
- **This presentation is consistent with acute adrenal insufficiency (adrenal crisis) from abrupt glucocorticoid withdrawal after HPA-axis suppression.** S27
- **Management requires immediate IV hydrocortisone (which has both glucocorticoid and mineralocorticoid activity) and IV fluids — this is a medical emergency, not a routine medication-refill issue.** S27; S27 (notes)

- The fix for a missed refill here is not "just restart the pill" — it is emergent IV steroid replacement; tapering protects against it. S27 (notes)
- Mifepristone's dual identity (glucocorticoid-receptor antagonist here vs. progesterone-receptor antagonist elsewhere) is worth flagging explicitly — don't conflate the two uses or assume a single mechanism. S23 (notes)
- Drug names reveal the class: **-tide** somatostatin analog (octreotide, lanreotide; lowers GH release); **-pressin** vasopressin analog (desmopressin; central diabetes insipidus); **-vaptan** V2 antagonist (tolvaptan; SIADH). S26; S26 (notes)
- **-ergoline** dopamine agonist, ergot-derived (cabergoline; lowers prolactin); **-thyroxine** thyroid hormone (levothyroxine; replacement, titrate to TSH); **-imazole / -thiouracil** thionamide antithyroid (methimazole, propylthiouracil; blocks hormone synthesis). S26
- **-sone / -solone** corticosteroid (prednisone, dexamethasone; anti-inflammatory, taper); **-conazole** azole steroidogenesis inhibitor (ketoconazole; Cushing syndrome); **-pristone** receptor antagonist (mifepristone; glucocorticoid-receptor block); **-lactone** mineralocorticoid antagonist (spironolactone; hyperkalemia risk). S26
- **A stem is a memory aid, not a rule: where no shared stem exists, learn the class name instead.** S26
- Recap: **1.** The hypothalamic-pituitary-target gland axis (Figure 1) is a shared regulatory template across thyroid, adrenal, and gonadal systems — drug therapy can intervene at any level. S28
- **2.** Thyroid hormone synthesis has distinct, drug-targetable steps (Figure 6); PTU uniquely blocks peripheral T4-to-T3 conversion, making it preferred in thyroid storm and first-trimester pregnancy. S28
- **3.** Chronic exogenous glucocorticoid use suppresses the HPA axis (Figure 9) — abrupt withdrawal risks adrenal crisis, so chronic therapy is tapered, not stopped. S28
- **4.** Adrenocortical antagonists (ketoconazole, metyrapone, mifepristone, spironolactone) block cortisol or aldosterone synthesis/action at different points for Cushing syndrome and hyperaldosteronism. **5.** Session 2 (Thursday, November 12) continues with gonadal hormones, diabetes, and parathyroid disorders. S28; S28 (notes)
- **TB** Older synthesis inhibitors — aminoglutethimide and trilostane — and the adrenolytic mitotane are rarely used; mifepristone (RU-486) is the receptor antagonist. p. 752; p. 753; p. 754
- **TB** Synthesis inhibitors & glucocorticoid antagonists: metyrapone is the only adrenal inhibitor usable in pregnancy; abiraterone treats refractory prostate cancer; mifepristone half-life about 20 hours. p. 754

Step by step

Chronic steroid patient S20; S22; S27; p. 750

On steroids > 2 weeks → assume HPA suppression →

Stopping → taper, never abruptly →

Missed doses + hypotension, confusion → adrenal crisis →

IV hydrocortisone + IV fluids (emergency) →

Addison disease: glucocorticoid + fludrocortisone

Figure 8 · Glucocorticoid receptor signaling. Schematic (illustrative) — Figure 8. A steroid enters the cell, binds its cytoplasmic receptor, and the complex switches genes on and off in the nucleus; NF-kB-driven cytokines are repressed. S21; S21 (notes)

Figure 9 · Exogenous corticosteroids and HPA suppression. Original illustrative diagram (not from Katzung): chronic exogenous glucocorticoid suppresses CRH and ACTH via negative feedback; endogenous cortisol falls; abrupt withdrawal → adrenal insufficiency. Taper. S22; S22 (notes)

Figure 10 · Blocking steroid synthesis vs. blocking the receptor. Schematic (illustrative) — Figure 10. Left: enzyme blocks on the cortisol pathway (ketoconazole, metyrapone). Right: receptor blocks (mifepristone at the glucocorticoid receptor, spironolactone at the mineralocorticoid receptor). S24; S24 (notes)

Glucocorticoids vs. mineralocorticoids (S20)

	Glucocorticoids (prednisone, dexamethasone)	Mineralocorticoids (fludrocortisone)
Main effect	Anti-inflammatory, immunosuppressive	Renal sodium retention, potassium excretion
Use	Inflammatory/autoimmune disease in nearly every organ system	Addison disease replacement, with a glucocorticoid
Monitor	Glucose, bone, infection, mood; taper chronic use	Blood pressure and potassium

How to tell them apart. Sources: S20

Table 1 · Adrenal pharmacology at a glance

Agent	Category	Key teaching point
Prednisone, dexamethasone	Glucocorticoid	Anti-inflammatory; chronic use risks HPA suppression — taper, don’t stop abruptly
Fludrocortisone	Mineralocorticoid	Sodium retention/potassium excretion; replacement in Addison disease
Ketoconazole, metyrapone	Steroidogenesis inhibitor	Block cortisol synthesis at different steps; used in Cushing syndrome

Agent	Category	Key teaching point
Mifepristone	Glucocorticoid-receptor antagonist	Hyperglycemia in Cushing syndrome when surgery isn't an option

How to tell them apart. Sources: S25; S25 (notes)

Supporting drugs at a glance (S52)

Drug(s)	Why it matters here
Antipsychotics and metoclopramide	Raise prolactin by blocking dopamine
Calcium, iron and antacids	Block levothyroxine absorption; separate by 4 hours
Amiodarone and lithium	Can cause hypo- or hyperthyroidism
Estrogens	Raise thyroid-binding globulin, so levothyroxine needs rise
Rifampin, phenytoin, carbamazepine	Speed clearance of levothyroxine and steroids
Azole antifungals	Raise corticosteroid exposure through CYP3A4
ACE inhibitors, ARBs and potassium	Add to spironolactone hyperkalemia
Insulin and oral agents	Doses need adjusting during steroid therapy

How to tell them apart. Sources: S52; S52 (notes)

Corticosteroid profiles: formulation, dose, indications

Drug	Class	Formulation	Adult dose	Indications
Prednisone	**Glucocorticoid (prodrug of prednisolone)**	Tablet 1/2.5/5/10/20/50 mg; oral solution	5-60 mg PO daily depending on indication; taper after more than 3 weeks of therapy	Inflammatory and autoimmune disease, allergic reactions
Dexamethasone (Decadron)	**Long-acting glucocorticoid**	Tablet 0.5-6 mg; oral solution; IV 4/10 mg/mL	0.5-9 mg PO or IV daily by indication; cerebral edema 10 mg IV load, then 4 mg every 6 hours; thyroid storm 2 mg IV every 6 hours	Cerebral edema, antiemetic, severe inflammation, Cushing screening test
Hydrocortisone (Cortef, Solu-Cortef)	**Glucocorticoid with mineralocorticoid activity**	Tablet 5/10/20 mg; IV 100 mg vial	Replacement 15-25 mg/day PO in 2-3 divided doses; adrenal crisis 100 mg IV, then 200 mg over 24 hours	Adrenal insufficiency, adrenal crisis, inflammatory conditions

Drug	Class	Formulation	Adult dose	Indications
Fludrocortisone (Florinef)	**Mineralocorticoid**	Tablet 0.1 mg	0.05-0.2 mg PO once daily	Primary adrenal insufficiency (with a glucocorticoid), orthostatic hypotension

How to tell them apart. Sources: S44; S45; S46; S47

Corticosteroid profiles: adverse effects, contraindications, toxicity

Drug	Adverse effects	Contraindications	Toxicity / overdose
Prednisone	Hyperglycemia, insomnia, mood change, weight gain, infection, osteoporosis	Systemic fungal infection; live vaccines at high dose	Adrenal suppression and crisis on abrupt stopping; Cushingoid features, cataracts, avascular necrosis
Dexamethasone (Decadron)	Insomnia, hyperglycemia, mood change, immune suppression	Systemic fungal infection, hypersensitivity	Potency about 25-30 times hydrocortisone with no mineralocorticoid effect; adrenal suppression, psychosis
Hydrocortisone (Cortef, Solu-Cortef)	Hyperglycemia, fluid retention, mood change, infection risk	Systemic fungal infection	Excess causes Cushingoid features, hypertension, hypokalemia; stress-dose during illness and surgery
Fludrocortisone (Florinef)	Hypertension, edema, hypokalemia, headache	Systemic fungal infection; caution in heart failure	Hypokalemia, fluid overload, supine hypertension; monitor blood pressure and potassium

How to tell them apart. Sources: S44; S45; S46; S47

Adrenal antagonist profiles: formulation, dose, indications

Drug	Class	Formulation	Adult dose	Indications
Ketoconazole	**Azole antifungal and steroidogenesis inhibitor**	Tablet 200 mg	Cushing syndrome (off-label): start 200 mg PO BID, usual 400-1,200 mg/day divided	Cushing syndrome (off-label), fungal infections
Metyrapone (Metopirone)	**11-beta-hydroxylase inhibitor**	Capsule 250 mg	Cushing syndrome: start 250 mg PO TID-QID, titrate to cortisol (up to 6 g/day); diagnostic test: 30 mg/kg once at midnight	Cushing syndrome, diagnostic test of the HPA axis

Drug	Class	Formulation	Adult dose	Indications
Mifepristone (Korlym)	**Glucocorticoid-receptor antagonist**	Tablet 300 mg (Cushing); 200 mg (pregnancy termination)	Cushing hyperglycemia: 300 mg PO daily with a meal, titrate up to 1,200 mg/day	Hyperglycemia with Cushing syndrome when surgery is not an option
Spironolactone (Aldactone)	**Mineralocorticoid-receptor antagonist**	Tablet 25/50/100 mg	Hyperaldosteronism: 25-100 mg PO daily (up to 400 mg); heart failure 12.5-50 mg daily	Primary hyperaldosteronism, heart failure, resistant hypertension, edema, hirsutism (off-label)

How to tell them apart. Sources: S48; S49; S50; S51

Adrenal antagonist profiles: adverse effects, contraindications, toxicity

Drug	Adverse effects	Contraindications	Toxicity / overdose
Ketoconazole	Nausea, abdominal pain, headache, gynecomastia, adrenal insufficiency	Liver disease, QT prolongation or interacting QT drugs, pregnancy	Boxed hepatotoxicity and QT prolongation; strong CYP3A4 inhibitor; monitor LFTs and ECG
Metyrapone (Metopirone)	Nausea, dizziness, hypertension, edema, hirsutism	Adrenal insufficiency, hypersensitivity	Adrenal insufficiency, hypokalemia, hypertension from deoxycorticosterone, androgen excess
Mifepristone (Korlym)	Fatigue, nausea, headache, hypokalemia, edema, endometrial thickening	Pregnancy (boxed warning), simvastatin or lovastatin, need for long-term corticosteroids	Pregnancy loss; adrenal insufficiency (cortisol rises so labs mislead); hypokalemia
Spironolactone (Aldactone)	Hyperkalemia, gynecomastia, menstrual irregularity, dizziness	Hyperkalemia, Addison disease, severe renal impairment	Life-threatening hyperkalemia (with ACE inhibitors, ARBs, potassium); monitor potassium and creatinine

How to tell them apart. Sources: S48; S49; S50; S51

WHY IT HAPPENS

- Why IV hydrocortisone in adrenal crisis, not just prednisone? Hydrocortisone replaces both glucocorticoid and mineralocorticoid action — and the patient is in shock. S21; S27

HIGH-YIELD

- Chronic steroids suppress CRH/ACTH → taper; abrupt stop → adrenal crisis → IV hydrocortisone + fluids. S22; S27; SG U4-R8
- Addison disease needs a glucocorticoid plus fludrocortisone. S20

Cram sheet

Every term, mechanism, drug, and number. Hide the answers and quiz yourself; tap a blurred answer to peek.

Prompt	Answer
Axis rule	Hormone feeds back to suppress the levels above S5; SG U4-R1
Acromegaly drugs	Octreotide, lanreotide, pegvisomant (IGF-1), cabergoline S6; S7
Octreotide AEs	GI upset, gallstones, bradycardia S7
Prolactinoma	Cabergoline (twice weekly) or bromocriptine S8; S9
DI vs SIADH	Desmopressin (hyponatremia) · fluid restriction, tolvaptan S8; S10
Levothyroxine	Empty stomach; separate Ca/Fe/antacids 4 h; TSH 6–8 weeks S12; S13
Over-replacement	AF, bone loss S13
Methimazole vs PTU	First line · first trimester and storm S15
Thionamide + fever/sore throat	Stop, check CBC (agranulocytosis) S15 (notes)
Storm order	Beta-blocker + PTU → iodine ≥ 1 h later → steroids S16; S17
Steroid potency	Prednisone ~4×, dexamethasone ~25–30× hydrocortisone S21
Adrenal crisis	IV hydrocortisone + IV fluids S27
Addison disease	Glucocorticoid + fludrocortisone S20
Cushing drugs	Ketoconazole, metyrapone (synthesis); mifepristone (receptor) S23
T4 half-life	7 days → once daily p. 733
Agranulocytosis	0.1–0.5%, first 90 days p. 731
Iodide escape	2–8 weeks if given alone p. 732
Adrenal suppression	After > 2 weeks of steroids p. 750

Self-test

First, the lecture's own 0 knowledge checks — exactly as written. Then original PANCE-style questions. Tap an answer to reveal the explanation.

From the lecture

Original questions

Q1 · Clinical vignette · Pituitary: GH, Prolactin & ADH

A patient with acromegaly starts pegvisomant. Which lab best tracks the response?

- **A** IGF-1
- **B** Prolactin
- **C** Sodium
- **D** Cortisol
- **E** Serum GH

ANSWER: A

Pegvisomant blocks the GH receptor so IGF-1 falls; GH rises and is not used to monitor it (S7). S7

- **B.** Not the target (S9).
- **C.** ADH drugs (S10).
- **D.** Adrenal axis (S22).
- **E.** GH rises on pegvisomant (S7).

Q2 · Mechanism · Pituitary: GH, Prolactin & ADH

How does cabergoline treat a prolactinoma?

- **A** Blocks dopamine D2 receptors
- **B** D2 agonism restores dopamine's brake
- **C** Blocks the prolactin receptor
- **D** Inhibits thyroid hormone release
- **E** Blocks somatostatin receptors

ANSWER: B

Dopamine normally inhibits prolactin; D2 agonists restore that tone and often shrink the tumor (S8–S9; SG U4-R3). S8; S9; SG U4-R3

- **A.** That raises prolactin (S9).
- **C.** Not the mechanism (S9).
- **D.** Not related (S16).
- **E.** Not the mechanism (S7).

Q3 · Mechanism · Pituitary: GH, Prolactin & ADH

How do octreotide and lanreotide treat acromegaly?

- **A** They block IGF-1 receptors
- **B** They activate D2 receptors
- **C** They replace missing GH
- **D** They block the GH receptor
- **E** They lower GH release

ANSWER: E

Somatostatin analogs bind somatostatin receptors and suppress GH release (S6–S7). S6; S7; SG U4-R2

- **A.** Not the mechanism (S7).
- **B.** Dopamine agonists (S9).
- **C.** That is somatropin (S6).
- **D.** That is pegvisomant (S7).

Q4 · Clinical vignette · Pituitary: GH, Prolactin & ADH

A patient with central diabetes insipidus starts desmopressin. What is the main safety counseling?

- **A** Drink extra water all day
- **B** Check liver tests monthly
- **C** Avoid sun exposure
- **D** Limit fluids (hyponatremia)
- **E** Watch for hypernatremia

ANSWER: D

Desmopressin keeps free water; risk is hyponatremia — monitor sodium and limit fluids (S10). S8; S10; SG U4-R4

- **A.** Raises hyponatremia risk (S10).
- **B.** Tolvaptan (S10).
- **C.** Demeclocycline (S10).
- **E.** Opposite direction (S10).

Q5 · Clinical vignette · Pituitary: GH, Prolactin & ADH

A patient with SIADH stays hyponatremic despite fluid restriction. Tolvaptan is planned. Why start it in hospital?

- **A** It causes hyperkalemia
- **B** It must be given IV only
- **C** Risk of sodium overcorrection
- **D** It causes severe hypoglycemia
- **E** It causes bradycardia

ANSWER: C

Tolvaptan can overcorrect sodium (osmotic demyelination) and injure the liver; start in hospital (S10). S8; S10

- **A.** Not the taught risk (S10).
- **B.** Oral drug (S37).
- **D.** Not tolvaptan (S10).
- **E.** Octreotide (S7).

Q6 · Clinical vignette · Thyroid & Antithyroid Drugs

A woman taking levothyroxine with her calcium and iron each morning has a high TSH. Best advice?

- **A** Separate them by at least 4 hours
- **B** Stop calcium permanently
- **C** Double the levothyroxine dose
- **D** Switch to methimazole
- **E** Take it with breakfast instead

ANSWER: A

Calcium, iron and antacids reduce absorption — separate by at least 4 hours; take it 30–60 minutes before breakfast (S12–S13). S12; S13; SG U4-R5

- **B.** Separation is enough (S13).
- **C.** Fix the timing first (S12).
- **D.** Antithyroid drug (S15).
- **E.** Empty stomach (S12).

Q7 · Clinical vignette · Thyroid & Antithyroid Drugs

A woman in her first trimester has hyperthyroidism. Which drug is preferred?

- **A** Radioactive iodine
- **B** Levothyroxine
- **C** Potassium iodide alone
- **D** Propylthiouracil
- **E** Methimazole

ANSWER: D

PTU is preferred in the first trimester; methimazole is avoided then (teratogenic) (S15). S15; SG U4-R6

- **A.** Contraindicated in pregnancy (p. 732).
- **B.** For hypothyroidism (S12).
- **C.** Escape and fetal goiter risk (p. 732).
- **E.** Avoid in first trimester (S15).

Q8 · Clinical vignette · Thyroid & Antithyroid Drugs

Six weeks into methimazole, a patient has fever and a sore throat. What should happen?

- **A** Switch to PTU immediately
- **B** Continue and add an antibiotic
- **C** Reassure: a common cold
- **D** Double the dose
- **E** Stop it and check a CBC

ANSWER: E

Thionamides carry an agranulocytosis warning; fever or sore throat → stop and check a CBC (S15). S15

- **A.** Check the CBC first (S15).
- **B.** Check for agranulocytosis first (S15).
- **C.** Not reassurance (S15).
- **D.** Not the answer (S15).

Q9 · Mechanism · Thyroid & Antithyroid Drugs

Why is PTU preferred over methimazole in thyroid storm?

- **A** It is better absorbed orally
- **B** It has no liver toxicity
- **C** It also blocks T4→T3 conversion
- **D** It blocks iodide trapping faster
- **E** It is a beta-blocker

ANSWER: C

PTU blocks synthesis and conversion, a faster effect on active hormone (S14–S15). S14; S15

- **A.** Not the reason (S15).
- **B.** Greater liver risk (S15).
- **D.** Not the reason (S14).
- **E.** Propranolol (S16).

Q10 · Mechanism · Thyroid & Antithyroid Drugs

What is levothyroxine, and how does it act?

- **A** Synthetic T4, converted to T3
- **B** A TRH analog acting centrally
- **C** Synthetic T3 acting directly
- **D** An iodide supplement for synthesis
- **E** A TSH-receptor agonist drug

ANSWER: A

Levothyroxine is synthetic T4, converted peripherally to active T3, which binds nuclear receptors (S12–S13). S12; S13

- **B.** Not the mechanism (S12).
- **C.** That is liothyronine (S12).
- **D.** Not the mechanism (S12).
- **E.** Not the mechanism (S12).

Q11 · Clinical vignette · Thyroid & Antithyroid Drugs

An 80-year-old with coronary disease is newly hypothyroid. How should levothyroxine be started?

- **A** A loading dose then full dose
- **B** Full 1.6 mcg/kg/day at once
- **C** No treatment until TSH > 20
- **D** Low dose, titrated slowly
- **E** Liothyronine (T3) at a high dose

ANSWER: D

Older adults and cardiac patients start low and titrate slowly to avoid angina or arrhythmia (S12). S12; S13

- **A.** Not the approach (S12).
- **B.** Risks angina (S12).
- **C.** Not the approach (S12).
- **E.** Not taught (S12).

Q12 · Clinical vignette · Thyroid & Antithyroid Drugs

A hypothyroid woman on levothyroxine learns she is pregnant. What dose change does the textbook advise?

- **A** Increase by about 25–30%
- **B** Halve the dose
- **C** No change; recheck after birth
- **D** Stop it until delivery
- **E** Switch to methimazole

ANSWER: A

Many pregnant hypothyroid patients need about 25–30% more thyroxine; add two extra doses a week as soon as pregnancy is confirmed (p. 734). p. 733; p. 734

- **B.** Needs rise (p. 734).
- **C.** Monitor closely (p. 734).
- **D.** Fetal brain needs it (p. 733).
- **E.** Wrong direction (p. 734).

Q13 · Mechanism · Adrenocorticosteroids & Antagonists

Why must chronic glucocorticoid therapy be tapered rather than stopped?

- **A** Because steroids cause dependence
- **B** To prevent hyperkalemia
- **C** The suppressed HPA axis needs time to recover
- **D** Because the adrenal gland is destroyed
- **E** To prevent rebound inflammation only

ANSWER: C

Exogenous steroid suppresses CRH and ACTH; abrupt withdrawal leaves too little endogenous cortisol (S22; SG U4-R8). S22; SG U4-R8

- **A.** Not the mechanism (S22).
- **B.** Not the reason (S22).
- **D.** Suppressed, not destroyed (S22).
- **E.** Not the main reason (S22).

Q14 · Clinical vignette · Adrenocorticosteroids & Antagonists

A patient with primary adrenal insufficiency is on hydrocortisone but has low BP and high potassium. What should be added?

- **A** Dexamethasone
- **B** Potassium supplement
- **C** Spironolactone
- **D** Ketoconazole
- **E** Fludrocortisone

ANSWER: E

Addison disease needs mineralocorticoid replacement alongside a glucocorticoid (S20). S20; S47

- **A.** Little mineralocorticoid action (S21).
- **B.** Wrong direction (S20).
- **C.** Blocks mineralocorticoid action (S23).
- **D.** Blocks steroid synthesis (S23).

Q15 · Mechanism · Adrenocorticosteroids & Antagonists

What does fludrocortisone do at the kidney?

- **A** Suppresses inflammation only
- **B** Retains sodium, excretes potassium
- **C** Blocks the mineralocorticoid receptor
- **D** Blocks ADH at V2 receptors
- **E** Excretes sodium, retains potassium

ANSWER: B

Mineralocorticoids promote renal sodium retention and potassium excretion; monitor BP and potassium (S20). S20

- **A.** Glucocorticoid action (S20).
- **C.** That is spironolactone (S23).
- **D.** That is tolvaptan (S10).
- **E.** Reversed (S20).

Q16 · Mechanism · Adrenocorticosteroids & Antagonists

How do chronic exogenous glucocorticoids cause adrenal insufficiency on abrupt withdrawal?

- **A** They block cortisol synthesis enzymes
- **B** They block the glucocorticoid receptor
- **C** Feedback suppresses CRH and ACTH
- **D** They destroy the adrenal cortex
- **E** They raise ACTH until it is exhausted

ANSWER: C

Chronic steroid suppresses CRH/ACTH via negative feedback, so endogenous cortisol cannot resume quickly (S22; S28). S22; S28

- **A.** That is metyrapone (S23).
- **B.** That is mifepristone (S23).
- **D.** Suppression, not destruction (S22).
- **E.** ACTH falls (S22).

Q17 · Mechanism · Adrenocorticosteroids & Antagonists

Which drug lowers cortisol by blocking 11-β-hydroxylase?

- **A** Mifepristone
- **B** Octreotide
- **C** Fludrocortisone
- **D** Spironolactone
- **E** Metyrapone

ANSWER: E

Metyrapone blocks the final step of cortisol synthesis (S23–S24). S23; S24; SG U4-R9

- **A.** Blocks the receptor (S23).
- **B.** Somatostatin analog (S7).
- **C.** Mineralocorticoid agonist (S20).
- **D.** Blocks the mineralocorticoid receptor (S23).

Q18 · Clinical vignette · Adrenocorticosteroids & Antagonists

A patient with Cushing syndrome on mifepristone has persistently high cortisol levels. What explains this?

- **A** It blocks 11-β-hydroxylase
- **B** Receptor blockade leaves cortisol high
- **C** The drug has failed
- **D** Lab interference from ketoconazole
- **E** It increases ACTH secretion by itself

ANSWER: B

Mifepristone blocks the glucocorticoid receptor, so cortisol levels rise — monitor clinically (S24). S24

- **A.** Metyrapone (S23).
- **C.** Expected effect (S24).
- **D.** Not the reason (S24).
- **E.** Not the taught mechanism (S24).

Objectives checklist

Tick only what you could teach out loud. Saved on this device.

- [ ] **IO4.1** Identify major pharmacologic classes used in the management of endocrine conditions and describe their mechanisms of action.
- [ ] **IO4.2** Describe the indications, contraindications, adverse effects, and clinically significant interactions associated with medications used to treat endocrine conditions.
- [ ] **IO4.3** Compare pharmacologic agents within and across therapeutic classes to distinguish clinically relevant differences in efficacy, safety, and use.
- [ ] **IO4.4** Select appropriate pharmacologic therapy for common endocrine conditions based on clinical presentation and patient-specific factors.
- [ ] **IO4.5** Identify appropriate dosing, administration, and monitoring considerations for medications used in endocrine conditions.
- [ ] **IO4.6** Explain important medication-use and safety information to patients receiving pharmacotherapy for endocrine conditions.

- [ ] **SO1** Describe the hypothalamic-pituitary-target gland axis as a shared regulatory template, and identify where drug therapy intervenes at each level.
- [ ] **SO2** Identify the major agents used for growth hormone disorders, hyperprolactinemia, and disorders of antidiuretic hormone (diabetes insipidus, SIADH).
- [ ] **SO3** Describe the mechanisms, indications, and monitoring of thyroid hormone replacement and antithyroid drug therapy, including management of thyroid storm.
- [ ] **SO4** Compare glucocorticoid and mineralocorticoid pharmacology, and identify agents used to block adrenocortical hormone synthesis or action.
- [ ] **SO5** Identify dosing, administration, and monitoring considerations, and explain key medication-safety counseling points to patients, including HPA-axis suppression and the need to taper chronic glucocorticoids.